Arjun S.D

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SUMMARY

Python Developer with a First-Class MSc in Data Analytics and 2+ years in production-grade data engineering and applied ML. Skilled in **Python (NumPy, pandas,, scikit-learn)**, **SQL**, and **cloud (Snowflake/Azure),** with experience shipping predictive models, A/B tests, and real-time dashboards. Strengths in feature engineering, experiment design, model evaluation and MLOps basics. Comfortable translating business problems into deployable pipelines and explainable models that drive measurable impact.

EDUCATION

MSc in Big Data Analytics | Sheffield Hallam University, UK | 2022 - 2023

- **Modules:** Machine Learning and AI **(84%)**, Data Analysis **(83%)**, Big Data and Distributed Systems **(80%)**
- **Dissertation:** Analysing the factors affecting student attainment in the UK **(84%)**

B.Tech in Computer Science and Engineering | APJ Abdul Kalam Technological University, India | 2017 - 2021

- **Modules**: Software Development Lifecycle **(85%),** Data Structure & Algorithms **(80%),** OOP **(79%)**
- **Dissertation**: Suspicious human activity recognition using Openpose and CNN ML model **(89%)**

WORK EXPERIENCE

AI Developer | GNX Digital Solutions| May 2026 - Present | Kochi, India

- AI Telephony Voice Agent & IVR: Developed a real-time conversational AI voice agent using Tata Smartflo, Google ADK, Gemini 2.5 Flash Native Audio, and Live API. Implemented custom barge-in detection, VAD tuning, audio resampling, and echo cancellation for low-latency multilingual voice interactions. Deployed using Docker with automated CI/CD.
- AI-Powered CV Parser & JD Matching: Built a resume screening platform using Vertex AI Gemini, FastAPI, OCR, and PostgreSQL to extract candidate information and calculate job-fit scores. Optimized extraction and validation pipelines, reducing average CV processing time from ~38–50 seconds to ~18–20 seconds.
- Intelligent Job Recommender: Developed a hybrid recruitment recommendation engine using pgvector, CV/JD embeddings, cosine similarity, skills matching, and experience-based scoring. Implemented eligibility filtering, improved embedding quality, and resolved matching inaccuracies across education, experience, and technical skills.
- HRMS Data Synchronization & ETL Architecture: Architected and implemented an end-to-end ETL pipeline to migrate and synchronize 30,000+ employee records from an external database to PostgreSQL. Replaced the legacy Node.js-based synchronization architecture, reducing processing time from 4 hours to under 20 minutes (92% faster).
- Distributed HRMS Synchronization: Engineered fault-tolerant asynchronous data synchronization using Celery, RabbitMQ, and Redis, implementing persistent messaging, lost-chunk recovery, retry mechanisms, and connection pooling. Developed resumable, duplicate-safe bulk vacancy uploads with background processing and configurable concurrency.
- AI Voice Intelligence: Developed a secondary AI agent for post-call transcript analysis to identify interested candidates missed by the live voice agent. Improved call outcome accuracy, eliminated conflicting portal submissions, and resolved telephony CDR synchronization race conditions.
- HRMS Integration & Workflow Automation: Enhanced employee data ingestion, dynamic language assignment, organizational hierarchy migrations, and HRMS API integrations. Improved data integrity across upload batches and streamlined administrative workflows.

AI Developer Intern | Zecser Business LLP| April 2026 - May 2026| Kochi, India

- Designed RESTful APIs and microservices using **FastAPI** and Python, with **Docker** containerization and CI/CD pipelines for deployment automation.
- Engineered a production-grade resume text extraction engine in **Python** supporting PDF and DOCX formats, with layout detection, noise cleaning, section detection, and confidence scoring
- Designed JSON schemas for candidate profiles and job descriptions across 4 core data entities with validation rules, versioning strategy.
- Created automated test suites using pytest covering PDF extraction, DOCX extraction, two-column layout detection, and section detection accuracy
- Developed **AI-powered video screening service** integrating behavioral analysis, real-time interview intelligence, and automated candidate scoring across multi-round hiring pipelines

Customer Service Advisor| Capita PLC | February 2025 - January 2026 | Coventry, UK

- Built an operations analytics suite in Power BI combining ticket data and knowledge-base usage to track AHT, FCR, backlog ageing, and deflection.
- Assisted in developing a data quality framework (regex/lookup checks, outlier flags, SLA rule tests) in **Python/SQL** added exceptions log + weekly quality scorecard to cut rework and re-opens.
- Built a daily SLA dashboard with alerts for tickets approaching breach (priority-weighted); enabled supervisors to reassign work and reduce last-minute escalations.
- Partnered with service leads to define KPIs and targets; wrote plain-English metric definitions and ensured one-click traceability from dashboard tiles to underlying records.

Python Developer | Infosys | July 2021 - September 2022 | Kerala, India

- Designed and developed ETL workflows to integrate heterogeneous sources into enterprise data warehouse, ensuring data completeness, accuracy, and timeliness.
- Delivered across the SDLC as Developer/Tester/Prod Support (L2/L3): planning, build, deployment, monitoring, and defect resolution within SLA using **Control-M** and **UNIX**.
- Wrote robust **Teradata** and **Snowflake SQL** (joins, window functions, CTEs, stored procs) for data integration and reporting; optimized queries to reduce runtime and compute cost.
- Utilized scripting languages (e.g.. **Python, SQL**) and **ETL tools** to automate data loading and testing process.
- Implemented **data quality checks** and automated validations, improving trust in downstream dashboards/reports.
- Built operational monitoring (job health, failure alerts, reruns) and handled incident/problem management, root-cause analysis, and permanent fixes.
- Conducted training for team members and end-users; created SOPs and code/config documentation to support audits and handovers.

ACADEMIC PROJECTS

Student Attainment Dashboard | Sheffield Hallam University | 2023

- Processed 2M+ education records using Python.
- Built interactive Power BI dashboards so decision-makers could rapidly identify at-risk student groups and intervene earlier.
- This was effectively an “early warning / targeted support” model in the education space. The same logic applies in adaptation: use analytics to identify who is most exposed, and act before harm escalates.

Crime Rates and Real Estate Dashboard | Sheffield Hallam University | 2023

- Built an **ETL pipeline** integrating **30M+ crime and property records** using **R**, **Hive** and **NiFi** for large-scale analytics.
- Applied **statistical and ML techniques** (regression, clustering, ANOVA) to explore links between crime rates, property values, and demographics.
- Designed interactive dashboards using **Tableau** for regional drill-downs, hypothesis testing, and insight visualisation.
- Ensured data governance & reproducibility with **Git versioning**, documented assumptions, and confidentiality controls.

CodeDoc AI Agent – LLM-powered assistant for code documentation

- Built an **AI-driven documentation generator** using **LLMs (LangChain + embeddings)** to convert code into structured markdown with summaries and function explanations.
- Built RESTful APIs with **FastAPI** backend, containerized with Docker, and deployed via CI/CD pipelines for production reliability.
- Designed an **evaluation workflow** (BLEU & readability scoring) and JSON-only output schema to ensure clarity, accuracy, and reproducibility.

ScamGuardAI – Spam/Fraud Email Classifier with LLM Risk Scoring

- Built a **hybrid ML + LLM** fraud triage app (**FastAPI** backend, **Streamlit UI)** that outputs a **0–100 risk score** with **plain-English rationale** and source snippets.
- Engineered features from **sender/URL heuristics + n-grams** and aggregated with **LLM judgments**; added **guardrails** and ML-only **fallback** for reliability.
- Implemented an **evaluation harness** (train/test, **precision/recall/F1**, ROC-AUC) with **cost-sensitive thresholds** to reduce false positives on legit mail.

Suspicious Human Activity Recognition | IEEE Publication | January 2021 – June 2021

- Conducted a deep learning–based research project to classify human activity as suspicious
- Designed and implemented a **Convolutional Neural Network (CNN)** architecture integrated with **2D pose estimation** to recognize actions.
- **Achieved 94.6% classification accuracy**, outperforming traditional motion-based techniques
- Used **Python, OpenCV, TensorFlow/Keras**, and **Matplotlib**

PUBLICATIONS

- **Dileep, A. S.,** Nabilah, S. S., Sreeju, S., Farhana, K., & Surumy, S**. (2022, March). Suspicious human activity recognition using 2D pose estimation and convolutional neural network.** In *2022 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET)* (pp. 19-23). **IEEE.** https://doi.org/10.1109/WiSPNET54241.2022.9767152
- **Book: Arjun, S. D.,** Al-Tamimi, A.-.K. (2025). **Education's Hidden Drivers: A Comprehensive Study of Academic Performance Drivers in English Universities**. LAP Lambert Academic Publishing.

EXTRACURRICULAR ACTIVITIES

- **Programming Coach** - Mentored **30+** Bachelor students to successfully pass coding exams

LANGUAGES, SKILLS, AND CERTIFICATES

- **Languages:** English (Professional), Malayalam (Native)
- **Programming & Data:** Python (pandas, NumPy, scikit-learn), SQL (Teradata/BTEQ, Snowflake, Postgres), R
- **Data Platforms & Pipelines:** ETL/ELT, Teradata, **Snowflake** (clustering, micro-partitions), UNIX, **Control-M**
- **Analytics & ML:** EDA, feature engineering, **A/B testing**, regression/classification
- **BI & Visualisation: Power BI** (DAX), decision-ready dashboards
- **Quality & Governance: Data lineage & S2T**, validation/reconciliation, auditability
- **Engineering Practice: Git/GitHub,** PR reviews, CI, pytest
- **GenAI/RAG & Agents: LangChain, LlamaIndex, FAISS, LangGraph, FastAPI, Streamlit, LLMOps (monitoring, eval)**
- **Certifications:** AWS Certified Cloud Practitioner(2024), AZ900 - MS Azure Fundamentals(2022)